The One Price Is the Quadratic

a(1/2) = -ln J_act at every leverage - the Gaussian identity F'(1/2) = F(1), universal

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L.O.R.E. Framework (Law of Repulsive Emanation)

Abstract

Applying known reality to Ch.82/83 yields a correction, not just a continuation. The universal "one price" of the reversible-mirror strand is not Delta mu = -ln J_act (which held only in the mild limit) but a(1/2) = -ln J_act, which holds at EVERY leverage, for the control and the strongest harvest coin alike, to +/-0.0004. It is exact because the overdamped trap produces near-Gaussian work (a known result for driven harmonic traps), so the scaled cumulant generating function is quadratic, F(t) = -mu t + (1/2) sigma^2 t^2, and the midpoint slope equals the endpoint value: F'(1/2) = F(1). In tilt variables this is a(1/2) = -ln J_act. The earlier Delta mu = -ln J_act is its mild-limit shadow: Delta mu subtracts the control derivative, which cancels only when J_control ~ 1 (reversibility). The quantity that grew monotonically in Ch.83/84 is NOT the failure of the price but the separation of Delta mu (the t=0 derivative) from the quadratic midpoint - exactly the irreversibility the feedback injects.

1. The Known Reality

For overdamped Brownian motion in a time-varying HARMONIC trap, the work is Gaussian (the state is an Ornstein-Uhlenbeck process; the work is a quadratic functional of it). Then the scaled cumulant generating function is QUADRATIC:

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F(t) = ln<E[e^{-tW}]> = -mu t + (1/2) sigma^2 t^2
F'(t) = -mu + sigma^2 t
=>  F'(1/2) = -mu + sigma^2/2 = F(1)
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The midpoint slope equals the endpoint value - a property of the parabola alone, needing no reversibility and no control reference. In tilt variables a(t) = -F'(t), J = E[e^{-W}] = e^{F(1)}:

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a(1/2) = -ln J_act        (EXACT for Gaussian work)
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2. Measured Universal

Over the whole frontier + control (signal mode), the same trap, seed 42, Heun SRK2:

row	a(1/2)	-lnJ	a + lnJ
control 0.5/0.5	-0.00156	-0.00159	+0.00003
eng 0.35/2	-0.07292	-0.07252	-0.00040
eng 0.25/4	-0.09635	-0.09599	-0.00036
eng 0.15/6	-0.11050	-0.11030	-0.00020
eng 0.10/8	-0.12175	-0.12145	-0.00031
harv 0.05/16	-0.12803	-0.12768	-0.00035

|a(1/2) + ln J_act| <= 0.0004 on every row - sampling level. The quadratic fit rms is ~8e-5 on every row, confirming the underlying F is parabolic to 1e-4. The identity is UNIVERSAL because it is the Gaussian quadratic structure, not a reversibility statement.

3. The Correction of Ch.82/83

The earlier "two ledgers one price" paired Delta mu with -ln J_act. Expanding:

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Delta mu = -F_c'(0) + F_0'(0)
Delta mu + ln J = F_c(1) - F_c'(0) + F_0'(0)
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This cancels to zero only when the CONTROL derivative F_0'(0) is negligible - i.e. when J_control ~ 1 (reversibility), the mild-coin regime. The UNIVERSAL identity a(1/2) = -ln J_act needs no control and no reversibility; and numerically a + ln J stays ~0 while Delta mu + ln J grows -0.024 at harvest. So:

4. The Deviation Is the Irreversibility, Not a Broken Price

The measured monotone |Delta mu + ln J| of Ch.83/84 is NOT a failure of the coin's price. It is Delta mu - the t = 0 derivative - leaving the quadratic midpoint as the feedback injects irreversibility. The price itself (a(1/2) = -ln J) never breaks:

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harv 0.05/16 :  a + ln J = -0.00035  yet  Delta mu + ln J = -0.024
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Main Results

THEOREM

Universal Price (Gaussian identity): on the trapped work, a(1/2) = -ln J_act exactly, because F is quadratic (near-Gaussian work) and F'(1/2) = F(1). Verified to +/-0.0004 on the control and every coin leverage, with F-quadratic rms ~8e-5.

COROLLARY

The Mild-Limit Shadow: Delta mu = -ln J_act of Ch.82/83 holds only when the control derivative F_0'(0) is negligible (J_control ~ 1, reversibility); it is the same price viewed from the origin, not a competing law.

COROLLARY

The Deviation Is Irreversibility: |Delta mu + ln J| grows (0.003 -> 0.024) as Delta mu leaves the quadratic midpoint, exactly the irreversibility the feedback injects; the price a(1/2) = -ln J never breaks.

Conclusion

Known reality corrects the record: the one price of the reversible-mirror strand is the quadratic identity a(1/2) = -ln J_act, universal because the trap's work is Gaussian. What Ch.83/84 saw growing is Delta mu leaving the quadratic midpoint - the feedback's irreversibility - not a broken law.

References

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